

# Scalable Probit Calibration

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## Abstract

Multinomial probit allows unrestricted substitution between alternatives, but its choice probabilities have no closed form and the standard simulation method computes them one alternative at a time. For the factor-structured case  $\Sigma = VV^\top + \text{diag}(D)$  we give an algorithm that computes all  $N$  probabilities in a single  $O(QNL)$  pass ( $Q$  factor nodes,  $L$  lattice points): conditionally on the factors the alternatives are independent, and a survival-product identity yields every alternative’s integral from one shared field. The map is reproducible and differentiable, with  $O(QNL)$  Jacobian–vector products, and its inverse — calibrating utilities to observed shares — is globally well-posed and solved in under a minute at  $N = 1000$ ; forward share errors stay below  $10^{-3}$  from  $N = 5$  to 5000 (21 seconds at  $N = 5000$ ). The method requires an adequate factor approximation of  $\Sigma$ ; benchmark and boundary experiments locate where it wins and where simulation remains the tool.

## 1 Introduction

When a consumer chooses one product from a large assortment, a commuter one route from a network, or a platform ranks one item above thousands of others, the question a model must answer is: if an alternative disappears or changes, *where does its demand go?* The tools used for this, essentially universally, are the logit family — multinomial logit, nested logit, and above all random-coefficients (mixed) logit in the BLP tradition, the workhorse of demand estimation in empirical industrial organization, antitrust, and pricing [11, 4, 12]. They are universal for one reason: they are the ones that can be computed. Choice modeling did not start there. It started with Thurstone’s 1927 model [1] — jointly Gaussian utilities, unrestricted correlation, hence unrestricted substitution — known in econometrics as multinomial probit (MNP).

Its probabilities are  $(N - 1)$ -dimensional Gaussian integrals with no closed form. The standard evaluator is the Geweke–Hajivassiliou–Keane (GHK) simulator [2, 3, 4]: it writes the probability that alternative  $i$  beats the rest as an orthant probability of the difference utilities and estimates it by sampling truncated normals sequentially along a Cholesky factor — a separate simulation *per alternative*, with  $R^{-1/2}$  error in the number of draws  $R$ . That cost is why probit survives mainly in small- $N$  settings (transportation panels, a handful of alternatives) and

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\*Numbers in this paper are produced by committed, seeded scripts with correctness anchors run before any comparison (<https://github.com/microprediction/kinetics>, experiments 13–18; the exact commit is recorded in the repository README and an immutable tag is pinned at circulation). The algorithm ships in the `thurstone` package; package–benchmark agreement is itself a reported number. Benchmarks: Apple M4, 16 GB, Python 3.12.9, NumPy 2.4.6, SciPy 1.18.0; experiment 17 enforces single-threaded BLAS via `threadpoolctl` and reports median-of-3 timings for sub-second calls; long simulations are single realizations, disclosed as such. Monte Carlo chunk sizes derive from an explicit memory budget (1.5 GB); peak resident memory stays below 2 GB.

effectively vanished from large- $N$  practice: the field kept the computable models and gave up the model it started from.

The concession is not free. Logit-family substitution is structurally restricted, and the whole apparatus of share inversion that demand estimation is built on exists on the logit side only. In the substitution study of Section 3, plain logit misallocates about 23% of the demand released by a deleted alternative where the correctly structured probit model misallocates 5%.

This paper’s point is that for the covariance structure applied work most often actually has — a factor model,  $\Sigma = VV^\top + \text{diag}(D)$  — this barrier is not real. We give one algorithm that solves *both directions* of the probit share map, plus its derivatives and counterfactuals:

1. **Forward pass** (utilities  $\rightarrow$  shares): all  $N$  choice probabilities at once in  $O(QNL)$  for  $Q$  factor nodes and  $L$  lattice points. All 5000 shares of an  $N = 5000$  problem in 21 seconds, at accuracy near the  $10^{-4}$  noise floor of the simulation reference; matching that with our per-alternative GHK implementation extrapolates to roughly fifty hours.
2. **Calibration** (observed shares  $\rightarrow$  utilities): the probit analogue of the logit-side inversions used throughout demand estimation [11, 12]. Globally well-posed (Theorem 1: existence and uniqueness modulo location) and solved for  $N = 1000$  in under a minute.
3. **Derivatives**: Jacobian–vector products in  $O(QNL)$  with no resampling noise; the Jacobian is an exact weighted graph Laplacian, so reduced systems are symmetric positive definite and Krylov-solvable.
4. **Counterfactuals**: all  $N$  single-removal share vectors from one conditional field pass in  $O(QN^2L)$ .

The mechanism is a single trick used four ways: conditionally on the  $k$  factors the alternatives are independent, and a lattice survival-product identity computes every alternative’s integral in one shared  $O(NL)$  pass — the field’s survival product is built once and each alternative’s leave-one-out integrand is recovered by division.

**Prior art.** The nearest methods, and what each lacks:

| approach                       | all $N$ shares                      | calibration                |
|--------------------------------|-------------------------------------|----------------------------|
| GHK simulation [2, 3]          | $N$ separate runs, $R^{-1/2}$ error | not attempted at scale     |
| analytic approximations [7, 8] | per-alternative                     | no                         |
| factor quadrature [5, 6]       | still per-alternative               | no                         |
| Bayesian factor MNP [17]       | simulation inside estimation        | posterior, not a share map |
| neural amortization [14]       | fast, uncontrolled error            | no                         |
| BLP-side inversion [11, 12]    | —                                   | logit family only          |
| <b>this paper</b>              | one $O(QNL)$ pass                   | Thm. 1; <1 min at $N=1000$ |

Figure 1 shows the running-time consequence as the number of alternatives grows: the per-alternative simulator’s cost bends upward and crosses the lattice transform near  $N \approx 200$  *at three orders of magnitude more error* (Table 1); by  $N = 5000$  the lattice computes all shares in 21 seconds where matched-accuracy GHK extrapolates to roughly fifty hours.

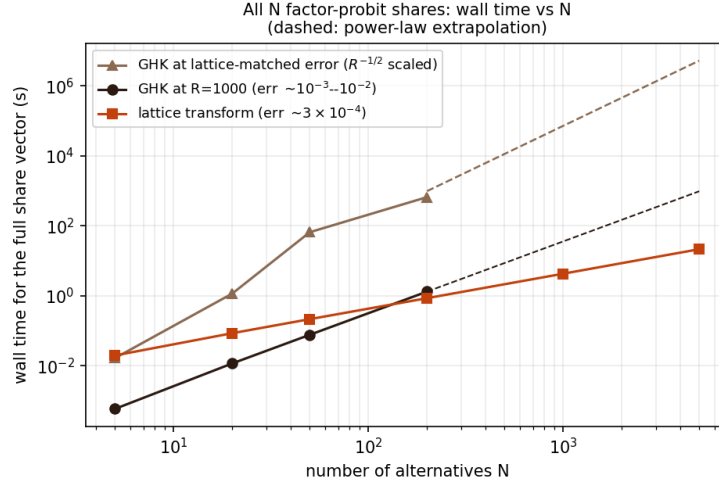


Figure 1: Wall time for the full  $N$ -alternative share vector as  $N$  grows ( $k = 2$  factors). The upper GHK line rescales  $R$  by the squared error ratio ( $R^{-1/2}$  scaling) so both methods deliver the same accuracy; at  $N = 5000$  it extrapolates to roughly fifty hours against the lattice’s 21 seconds (Section 2). Dashed segments are power-law extrapolations; all other points are measured (Table 1).

**Scope.** The method needs the factor form: where  $\Sigma$  has no adequate low-rank-plus-diagonal approximation in the choice-relevant quotient space, simulation at exact  $\Sigma$  remains the tool (Section 3). At coarse accuracy targets ( $\sim 10^{-3}$ ), plain direct utility simulation is a genuinely competitive forward baseline (Section 2); the decisive advantages are error decay below that level, reproducibility, resampling-free derivatives, and the inversion and counterfactual machinery.

The one-pass identity itself comes from an unlikely literature — inferring ability from win probabilities in multi-entrant races [10] — which is perhaps why it has not been assembled for probit before, despite each ingredient being classical.

**Model.** Throughout,

$$U_i = \mu_i + v_i^\top f + \sqrt{D_i} \varepsilon_i, \quad f \sim N(0, I_k), \quad \varepsilon_i \sim N(0, 1) \text{ i.i.d.}, \quad D_i > 0, \quad (1)$$

i.e.  $\Sigma = VV^\top + \text{diag}(D)$ ; the chosen alternative is the argmax of  $U$ . Factor dimension reduction for probit is classical — one-factor quadrature goes back to Butler and Moffitt [5], factor-analytic probit to Elrod and Keane [6]. The contribution is the all-alternatives assembly, the inverse transform, the derivative and counterfactual structure, and evidence for all of it.

**Proposition 1** (Conditional identity). *Under (1), with  $g_i(\cdot | f)$  and  $F_i(\cdot | f)$  the conditional density and CDF of  $U_i$  given  $f$ ,*

$$p_i = \mathbb{E}_f \left[ \int g_i(x | f) \prod_{j \neq i} F_j(x | f) dx \right], \quad \sum_i p_i = 1, \quad (2)$$

and the product over  $j \neq i$ , for all  $i$  simultaneously, is obtained from  $\log F_{\text{field}} = \sum_j \log F_j$  by subtraction.

*Proof.* Conditional on  $f$  the  $U_j$  are independent, so  $\mathbb{P}(U_i \text{ maximal} \mid f) = \int g_i \prod_{j \neq i} F_j dx$ ; average over  $f$ . The events partition (ties are null for continuous laws), giving  $\sum_i p_i = 1$ .  $\square$

**Proposition 2** (Complexity). *With  $Q$  factor nodes and  $L$  lattice points, the forward share vector costs  $O(QNL)$ ; the full single-removal ensemble  $\{P(j \text{ chosen} \mid i \text{ removed})\}_{i,j}$  costs  $O(QN^2L)$  arithmetic and  $O(N^2)$  storage, reusing one conditional field construction.*

*Proof.* Per node: the field log-product costs  $O(NL)$ , each alternative's integral  $O(L)$  after subtraction, giving  $O(NL)$ ; the removal ensemble evaluates  $N$  survivor integrals for each of  $N$  removals,  $O(N^2L)$ , over the same cached field.  $\square$

**Proposition 3** (Identification and differentiability).  *$p(\mu + c\mathbf{1}) = p(\mu)$ , so  $J = \partial p / \partial \mu$  has  $\mathbf{1}$  in its null space and  $\mu$  is identified only up to location. With  $B$  an  $N \times (N - 1)$  basis of  $\mathbf{1}^\perp$  and  $\mu = B\eta$ , share inversion and implicit differentiation are taken in reduced coordinates:  $d\eta/d\theta = -(B^\top JB)^{-1}B^\top \partial p / \partial \theta$ , provided  $B^\top JB$  is nonsingular.*

*Proof.* Invariance holds because a common location shift does not change the argmax; differentiating  $p(\mu + c\mathbf{1}) = p(\mu)$  in  $c$  gives  $J\mathbf{1} = 0$ . Differentiating  $p(B\eta(\theta), \theta) = \hat{p}$  in  $\theta$  and projecting with  $B^\top$  gives the reduced formula under the stated regularity.  $\square$

**Theorem 1** (Weighted-Laplacian structure and global inversion). *For model (1) with  $D_i > 0$ , in max-wins coordinates, define for  $i \neq j$*

$$w_{ij} = \mathbb{E}_f \int g_i(x \mid f) g_j(x \mid f) \prod_{\ell \neq i,j} F_\ell(x \mid f) dx > 0.$$

*Then  $J = \partial p / \partial \mu$  satisfies  $J_{ij} = -w_{ij}$  and  $J_{ii} = \sum_{j \neq i} w_{ij}$ , i.e.*

$$J = \sum_{i < j} w_{ij} (e_i - e_j)(e_i - e_j)^\top, \quad h^\top J h = \sum_{i < j} w_{ij} (h_i - h_j)^2,$$

*a weighted graph Laplacian with everywhere-positive weights:  $J$  is symmetric, positive semidefinite, has  $\mathbf{1}$  as its only null direction, and  $B^\top JB \succ 0$  for every full-rank basis  $B$  of  $\mathbf{1}^\perp$ . Moreover  $p = \nabla G$  for the social-surplus function  $G(\mu) = \mathbb{E} \max_i \{\mu_i + \xi_i\}$ , and the restriction of  $p$  to  $\mathbf{1}^\perp$  is a smooth bijection onto the interior of the probability simplex: every strictly positive target  $\hat{p}$  has exactly one mean-zero utility vector.*

*Proof.* Differentiating (2) in  $\mu_j$  and integrating by parts gives  $J_{ij} = -w_{ij}$  for  $i \neq j$ ; the diagonal follows from  $J\mathbf{1} = 0$  (Proposition 3), and positivity of  $w_{ij}$  from positivity of Gaussian densities. The quadratic form is then immediate, and it vanishes only for constant  $h$ . For inversion, minimize  $H(\mu) = G(\mu) - \hat{p}^\top \mu$  over  $\mathbf{1}^\perp$ :  $H$  is strictly convex there (its Hessian is the reduced  $J$ ), and since the shocks have zero mean,  $G(\mu) \geq \max_i \mu_i$ , so

$$H(\mu) \geq \max_i \mu_i - \hat{p}^\top \mu = \sum_i \hat{p}_i (\max_j \mu_j - \mu_i) \geq \hat{p}_{\min} (\max_i \mu_i - \min_i \mu_i) \rightarrow \infty$$

as  $\|\mu\| \rightarrow \infty$  on  $\mathbf{1}^\perp$ . The unique minimizer satisfies  $\nabla G = \hat{p}$ .  $\square$

*Remark 1.* Social-surplus convex duality for discrete choice is classical [15, 16]; what is new here is not well-posedness but its scalable realization. Each identity in the theorem is also checked numerically in the repository: the  $w_{ij}$  representation against finite differences to  $6 \times 10^{-9}$ , symmetry to  $4 \times 10^{-12}$ , and positive definiteness of the reduced numerical Jacobian.

**Proposition 4** (Jacobian–vector products in  $O(QNL)$ ). *With hazards  $\lambda_j = g_j/F_j$ ,  $\Lambda = \sum_j \lambda_j$ , and  $A = \sum_j h_j \lambda_j$  (all conditional on  $f$ , all functions of  $x$ ), the Jacobian of the exact max-wins map acts on a direction  $h$  as*

$$(Jh)_i = \mathbb{E}_f \int g_i \left( \prod_{j \neq i} F_j \right) (h_i \Lambda - A) dx,$$

where the field sums  $\Lambda$  and  $A$  are formed once per node and lattice point, so all  $N$  components cost  $O(NL)$  per node. Translation invariance is visible ( $h = \mathbf{1}$  gives  $A = \Lambda$ , hence  $Jh = 0$ ), and the weighted-Laplacian form of Theorem 1 follows by expanding  $A$ . This enables Krylov solves of the reduced system in Proposition 3 without forming  $J$  densely at  $O(QN^2L)$ .

*Proof.* Write  $R_i = \prod_{j \neq i} F_j$ . For the own derivative,  $\partial_{\mu_i} g_i(x) = -g'_i(x)$  (a location shift), and integrating by parts,  $\int (-g'_i) R_i dx = \int g_i R'_i dx = \int g_i R_i \sum_{j \neq i} \lambda_j dx = \int g_i R_i (\Lambda - \lambda_i) dx$ . For  $j \neq i$ ,  $\partial_{\mu_j} F_j = -g_j$ , so the cross derivative is  $-\int g_i g_j \prod_{\ell \neq i,j} F_\ell dx = -\int g_i R_i \lambda_j dx$ . Summing against  $h$ ,  $(Jh)_i = \int g_i R_i [h_i (\Lambda - \lambda_i) - \sum_{j \neq i} h_j \lambda_j] dx = \int g_i R_i (h_i \Lambda - A) dx$ ; average over  $f$ .  $\square$

**The differentiated map.** Three maps must be distinguished: the exact max-wins map  $p^{\max}(\mu)$ ; the reflected min-wins map  $p^{\min}(a)$  used by the implementation, related by  $a = -\mu$  so that  $D_\mu p^{\max}[h] = D_a p^{\min}[-h]$ ; and the implemented approximation  $p_{Q,L} = \tilde{p}_{Q,L} / (\mathbf{1}^\top \tilde{p}_{Q,L})$ , whose derivative carries a normalization correction: with  $v = D\tilde{p}[h]$  and  $s = \mathbf{1}^\top \tilde{p}$ ,

$$Dp_{Q,L}[h] = \frac{v - p_{Q,L} (\mathbf{1}^\top v)}{s}.$$

Since  $\mathbf{1}^\top v = 0$  analytically, the correction is of the order of the quadrature defect; empirically the uncorrected formula matches central finite differences of the returned normalized map to  $5 \times 10^{-9}$ . One further convention: the adaptive lattice endpoints depend on  $\mu$ , so derivative statements refer to the *fixed-design* map — the lattice envelope and factor nodes are frozen during any inversion or outer-derivative computation, and the inversion’s analytic own-location slopes are accordingly a Jacobi quasi-Newton preconditioner rather than the diagonal of the adaptive map’s Jacobian.

**Tail-stable implementation.** Tail stability requires the log domain throughout:  $\log \Phi$  evaluated directly (never  $1 - \Phi$ , which underflows in double precision at  $z \approx 8.3$ ), the analytic  $\log g$ , and hazards as  $\exp(\log g - \log \Phi)$ , which is bounded ( $\sim z$  in the tail). Flooring a density before taking its logarithm while the exact log-survival stays finite produces spurious  $e^{+92}$  hazards on problems as mild as  $N = 8$  with heterogeneous  $D$ ; that configuration is a regression test. In the forward map the same discipline lets deep-tail shares resolve to genuine small positives rather than floored zeros.

**Beyond Gaussian factors.** The factors need not be Gaussian: (2) requires only conditional independence and an integrable factor law, so Student, mixture, or empirical factor distributions are admissible with appropriate nodes, and the base densities may be arbitrary and alternative-specific. One caution for asymmetric laws: the implementation works in the reflected (min-wins) race, and  $-U$  has the same distribution as the reflected model only when the factor and base laws are symmetric; for asymmetric laws the reflected distributions must be used explicitly.

**Numerical method.** The benchmarks use  $L = 1501$  lattice points on the adaptive interval  $[\min_{i,q} m_i(f_q) - 8 \max_i \sqrt{D_i}, \max_{i,q} m_i(f_q) + 8 \max_i \sqrt{D_i}]$ , where  $q = 1, \dots, Q$  indexes the retained factor nodes.

Factor integration: the GHK benchmark (experiment 13) uses product Gauss–Hermite of order 15/11/9 for  $k = 2/3/4$ ; the boundary experiment (experiment 14) uses order 15 per dimension for all  $k \leq 4$ , which after pruning weights below  $10^{-7}$  of the maximum leaves  $Q = 145, 1317$ , and  $10929$  nodes for  $k = 2, 3, 4$  — the exponential growth of  $Q$  with rank made explicit. Scrambled-Sobol nodes ( $2^{13}$  points, fixed seed) are used for  $k > 4$ : reproducible fixed-seed randomized QMC, disclosed as such.

Survival floors sit at  $10^{-300}$  and the quadrature output is normalized by its sum. The returned map is a reproducible, piecewise-smooth numerical approximation of (2); statements about derivatives refer to derivatives of this approximation, whose fidelity to exact MNP derivatives is governed by  $L$  and  $Q$ .

**Algorithm.** Orientation: the implementation computes the reflected min-wins race on  $a = -\mu$ . Per retained factor node  $q$ : form conditional locations  $m_i(f_q) = a_i + v_i^\top f_q$ ; on the uniform lattice evaluate  $\log S_i = \log \Phi(-(x - m_i)/\sqrt{D_i})$ ; accumulate the field sum  $\log S_{\text{field}} = \sum_j \log S_j$ ; recover each alternative’s integrand as  $g_i \exp(\log S_{\text{field}} - \log S_i)$ ; integrate by the rectangle rule ( $\Delta x$  weights, no endpoint modification — for integrands decaying to zero at both ends this equals the trapezoidal rule). Pruned factor weights are not renormalized explicitly; the deficit is absorbed by the final normalization  $p = \tilde{p}/(\mathbf{1}^\top \tilde{p})$ , and  $|1 - \mathbf{1}^\top \tilde{p}|$  is reported as a diagnostic. Inversion: Newton step  $\mu \leftarrow \mu - \text{damp} \cdot r/\partial_i$ , with  $r$  the log-share residual on identified alternatives,  $\partial_i$  the analytic own-location log-slope, steps capped at  $\sqrt{D_i + \|v_i\|^2}$ , damping halved (floor 0.25) whenever the residual grows by more than 20%, tolerance  $10^{-6}$  on the identified set, at most 50 iterations, followed by mean removal.

**Inversion.** Given target shares  $\hat{p}$  and  $(V, D)$ , the inverse transform finds mean-zero  $\mu$  with  $p(\mu) = \hat{p}$  by a damped Jacobi quasi-Newton iteration: all coordinates updated simultaneously with own-location slopes of the unnormalized lattice map as the preconditioner (available from the same pass), the field refrozen each iteration, a warm start from the independent-field inverse using total variances, and a tail-aware tolerance (alternatives with target shares below  $10^{-4}/N$  are matched best-effort; their locations are weakly identified — itself a conditioning caution for the reduced Jacobian when shares are tiny or alternatives nearly collinear). Convergence in a handful of iterations is an empirical observation on the tested problems, not a theorem. The design follows the original transform’s inversion [10] and the calibration practice of [18].

## 2 Benchmark against GHK

**Correctness anchors.** On the binary case, where MNP has a closed form, the lattice transform agrees to  $3.3 \times 10^{-16}$  and our GHK implementation — validated before being compared against — to  $2 \times 10^{-14}$  at  $R = 2 \times 10^5$  (an easy anchor for both methods, useful for code validation rather than as evidence about the hard regime). At  $N = 5$  both methods sit within the noise of a  $10^7$ -draw Monte Carlo reference, and the shipped `thurstone.FactorRace` agrees with the benchmark kernel to  $1.1 \times 10^{-5}$ .

**Error metric.** Throughout,  $\max_i |\hat{p}_i - p_i|$  against a simulation reference. A maximum over  $N$  coordinates tends to grow with  $N$  at fixed per-coordinate accuracy, and the reference’s own maximum-coordinate noise behaves likewise; mean-absolute and total-variation summaries are committed alongside (experiment 16): at  $N = 1000$  the lattice’s mean absolute error is  $5.6 \times 10^{-6}$  and total variation  $2.8 \times 10^{-3}$  against a fresh  $2 \times 10^6$ -draw reference.

| $N$  | lattice transform                        | GHK ( $R = 1000$ )                |
|------|------------------------------------------|-----------------------------------|
| 5    | 20 ms, err $5.7 \times 10^{-4}$          | 1 ms, err $3.1 \times 10^{-3}$    |
| 20   | 86 ms, err $4.4 \times 10^{-4}$          | 12 ms, err $4.4 \times 10^{-3}$   |
| 50   | 217 ms, err $2.7 \times 10^{-4}$         | 77 ms, err $7.9 \times 10^{-3}$   |
| 200  | 860 ms, err $3.0 \times 10^{-4}$         | 1323 ms, err $6.8 \times 10^{-3}$ |
| 1000 | 4.3 s, err $6.8 \times 10^{-4} \dagger$  | —                                 |
| 5000 | 21.5 s, err $5.0 \times 10^{-4} \dagger$ | —                                 |

Table 1: Full share vector,  $k = 2$  factors, one problem per size, generated as  $\mu_i \sim N(0, s^2)$  with spread  $s = 1$  through  $N = 200$  and 1.5 above,  $v_{ir} \sim N(0, 0.25/k)$ ,  $D_i \sim U[0.5, 1.5]$ . Sub-second timings are warmed medians of three runs. Reference:  $2 \times 10^6$  MC draws through  $N = 200$ ;  $5 \times 10^5$  draws at  $N = 1000$  and 5000 ( $\dagger$ : there the reference’s own maximum-coordinate noise is of comparable order, so these entries bound the lattice error only to that order). A replication study at common spread 1.0 (experiment 16; ten problems per size at  $N \leq 200$ , three at  $N = 1000$ , twin independent  $2 \times 10^6$ -draw references per problem) gives worst-case lattice error  $3.5\text{--}4.6 \times 10^{-4}$  at every size, at or below the references’ own replicate-to-replicate noise. Lattice error stayed below  $10^{-3}$  throughout; GHK cost crossed over by  $N \approx 200$ .

**Matched-accuracy extrapolation.** Fitting a power law to the reference implementation’s timings over  $N \in [20, 200]$  ( $\alpha \approx 2.0$ ; a lower bound, the asymptotic per-share cost being cubic) and combining with  $R^{-1/2}$  error scaling suggests that matching the lattice’s  $5 \times 10^{-4}$  at  $N = 5000$  would take this implementation on the order of fifty hours (0.27 h at  $R = 1000$ , times  $(6.8 \times 10^{-3}/5 \times 10^{-4})^2 \approx 185$ ), against 21 seconds. The baseline is a straightforward single-threaded Python GHK: per-alternative Cholesky, plain pseudorandom uniforms, one realization per problem, shares normalized by their sum. It establishes the scaling of the per-alternative simulation approach, not a field-wide impossibility.

**Direct simulation baseline.** The strongest cheap baseline is *direct utility simulation* — draw utilities, take argmax, average — which produces the whole share vector per draw and, unlike GHK, does not pay per alternative. We ran it at wall time matched to the lattice call (experiment 16): its maximum-coordinate error was 1.5–4 $\times$  the lattice’s at every tested  $N$  ( $6.2 \times 10^{-4}$  versus  $3.9 \times 10^{-4}$  at  $N = 1000$ , 0.5M draws in the matched 4.1 seconds). At the  $10^{-3}$  accuracy level, then, raw forward speed does not by itself separate the methods.

The separation is in error decay (experiment 17,  $N = 200$ ,  $k = 2$ ). Holding the factor rule fixed, lattice self-convergence is  $2.7 \times 10^{-8}$  at  $L = 101$  and at machine precision from  $L \approx 200$  — the integrand is analytic and endpoint-decaying, so the uniform lattice is effectively spectrally accurate and  $L$ -error is subdominant. Holding  $L$  fixed, the Gauss–Hermite sweep decays from  $5.3 \times 10^{-4}$  (order 3) to  $5.9 \times 10^{-9}$  (order 15). The pre-normalization defect  $|1 - \mathbf{1}^\top \tilde{p}|$  is  $2.4 \times 10^{-8}$ .

On the same problem the measured accuracy–time frontier (Figure 2, left) has the lattice

at  $4.1 \times 10^{-5}$  error in 0.03 seconds (at the twin-reference noise floor), direct MC at  $1.2 \times 10^{-4}$  in 15.7 seconds ( $R = 10^7$ ), scrambled-Sobol direct QMC at  $2.0 \times 10^{-4}$  in 2.6 seconds ( $2^{20}$  points), GHK at  $2.0 \times 10^{-3}$  in 10.7 seconds ( $R = 10^4$ ; over 8 seeds at  $R = 10^3$  its error spans  $[3.9 \times 10^{-3}, 1.3 \times 10^{-2}]$ ), and QMC-GHK (scrambled-Sobol uniforms in the same kernel) at  $4.2 \times 10^{-4}$  in 7.0 seconds ( $R = 8192$ ) — a 5–7 $\times$  error improvement over plain GHK at equal  $R$ , still well off the lattice frontier. Below  $10^{-3}$  the deterministic map separates decisively, consistent with  $R^{-1/2}$  against a rapidly-converging quadrature. One stronger stochastic baseline — minimax tilting [9] — remains to be added.

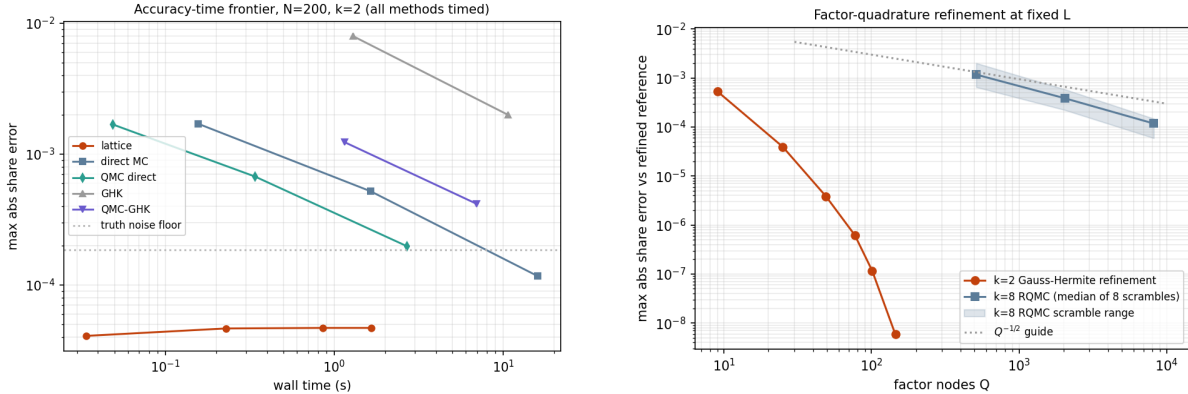


Figure 2: Left: measured accuracy–time frontier at  $N = 200$ ,  $k = 2$  (experiment 17) — every method timed on the same problem against twin  $5 \times 10^7$ -draw references; direct simulation and QMC-GHK are first-class baselines. Right: factor-quadrature refinement at fixed  $L$ : Gauss–Hermite at  $k = 2$ , and at  $k = 8$  the median and range over eight independent RQMC scrambles.

**Derivatives.** For fixed factor nodes and a fixed lattice, the transform is a reproducible, (piecewise-)differentiable function of  $\mu$  whose derivatives are evaluated without resampling; analytic own-location slopes of the unnormalized map come from the same pass and serve as the inversion preconditioner. (Along a utility path at  $N = 50$ , the maximum second difference over  $dt^2$  is 39.3 for fresh-draw GHK against 0.01 for both CRN-GHK and the lattice; the committed smoothness figure shows the curves.) These are derivatives of the numerical approximation, not exact MNP derivatives, and GHK-based practice likewise differentiates its approximation, analytically or with fixed draws [3, 4]. The practical distinction is the character and controllability of the approximation error — quadrature resolution versus draw count — not the possibility of differentiation.

**Share inversion.** At  $N = 1000$ ,  $k = 2$ , with targets from  $5 \times 10^6$  MC draws floored at  $10^{-7}$  and renormalized: the inversion completes in 57 seconds. The iteration solves its own equations to  $6.4 \times 10^{-9}$  — a statement about the solver, not statistical accuracy, since the target carries sampling noise  $\sim 2 \times 10^{-4}$  — and recovers the utilities of identified alternatives (target share above  $3 \times 10^{-4}$ ) to maximum error 0.017. A replication study (experiment 16) repeats the inversion on a second  $N = 1000$  problem against three independent  $5 \times 10^6$ -draw targets: 57 seconds each, 161–162 alternatives identified (98.1% of share mass), recovery error maximum 0.015–0.023 and median  $\approx 0.0027$  across replicates, with the recovery-versus-share



curve committed alongside. Numerical well-posedness matches Theorem 1: the reduced Jacobian assembled from  $N$  JVP calls at  $N = 50$  is symmetric to  $1.7 \times 10^{-18}$ , has row-sum defect  $6.2 \times 10^{-17}$ , and reduced eigenvalues in  $[2.5 \times 10^{-5}, 0.22]$  — positive definite as proved. We are not aware of probit share inversion at this scale in current simulation-based practice.

**Removal ensemble.** All 200 single-removal share vectors come from one conditional field construction in  $O(QN^2L)$ : 51.5 seconds versus 172.4 seconds recomputing per removal, with ten removals checked directly (agreement  $2.8 \times 10^{-17}$  — an algebraic-consistency check between two implementations of the same approximation, not an accuracy statement). The strongest simulation comparator also reuses its draws: one draw’s winner  $w$  and runner-up  $s$  determine every single-removal outcome (delete  $i \neq w$ : winner stays  $w$ ; delete  $w$ : winner becomes  $s$ ), so counting winners and winner/runner-up pairs yields the whole  $N \times N$  deletion matrix in  $O(RN + N^2)$ . At wall time matched to the field ensemble (64 s), this top-two baseline reaches  $1.2 \times 10^{-4}$  maximum error on the twenty highest-share deletion rows against an independent  $10^8$ -draw reference, versus  $4.2 \times 10^{-5}$  for the conditional field (experiment 18): the deterministic ensemble is about  $3 \times$  more accurate at matched time, and the reuse trick, not the field construction alone, is the right simulation benchmark.

### 3 Boundary experiments

**General covariance through the factor approximation.** The method requires  $\Sigma \approx VV^\top + D$ ; GHK does not. Choices depend on  $(\mu, \Sigma)$  only through  $P\mu$  and  $P\Sigma P$ ,  $P = I - \mathbf{1}\mathbf{1}^\top/N$ : a common factor loading  $V \rightarrow V + \mathbf{1}c^\top$  shifts every conditional location equally and cannot move the argmax (a machine-precision identity), so the factor fit must be run in this choice-relevant quotient space — fitting raw  $\Sigma$  can spend scarce rank on an irrelevant common component ( $\Sigma = \tau^2 \mathbf{1}\mathbf{1}^\top + \beta\beta^\top + D$  with large  $\tau$ : the raw rank-1 fit takes  $\mathbf{1}\mathbf{1}^\top$  and misses  $\beta\beta^\top$  entirely; on such an example the contrast-space fit cuts rank-1 share error from  $4.6 \times 10^{-2}$  to  $8.3 \times 10^{-3}$ ). The asymmetry matters: only the common *factor* direction is irrelevant; a common addition to  $D$  inflates every difference variance and is choice-relevant, so  $D$  must come from the quotient fit. Loadings are centered and SVD-canonicalized (ordered columns, fixed signs), making results reproducible at the covariance level.

With this fit: correlation matrices at spectral decay  $\lambda_m \propto m^{-\gamma}$ ,  $\gamma \in \{0.5, 1.5, 3\}$ , at  $N = 50$ , sharing one orthogonal eigenbasis across  $\gamma$  so decay comparisons are not confounded with basis realization. The power law holds before diagonal standardization, which perturbs the spectrum; the actual leading eigenvalues (3.7, 17.1, 34.2 at  $\gamma = 0.5, 1.5, 3$ ) are printed by the script. Share error is measured against an  $8 \times 10^6$ -draw reference, with GHK at the exact  $\Sigma$  for comparison; the plotted quantity is the rank- $k$  approximation error of the stated procedure, not a minimum over factor models. Rank-8 errors:  $1.8 \times 10^{-3}$ ,  $3.7 \times 10^{-3}$ , and  $3.1 \times 10^{-3}$  at  $\gamma = 0.5, 1.5, 3$  from  $k = 1$  starting points of  $2.3 \times 10^{-3}$ ,  $1.5 \times 10^{-2}$ ,  $7.1 \times 10^{-2}$ .

Decomposing the error with a second simulation under the fitted factor model: integration error is flat at  $2\text{--}7 \times 10^{-4}$  across every  $(\gamma, k)$  tested, so the rank- $k$  error is essentially covariance-fit error — rank, not quadrature, is the binding constraint. Against GHK at  $R = 10^4$  (3.2, 2.3,  $4.3 \times 10^{-3}$ ), rank-8 wins at  $\gamma = 0.5$  and 3 but *loses at intermediate decay*  $\gamma = 1.5$  ( $3.7 \times 10^{-3}$  versus  $2.3 \times 10^{-3}$ ): a GHK-wins regime does exist, where the spectrum is too flat to truncate and too structured to ignore. Nor is this wall-time-matched: the  $k = 8$  RQMC evaluation takes 15–24 s against 0.5–0.9 s for GHK at  $R = 10^4$ , so at  $N = 50$  GHK is the cheaper route to

$3\text{--}4 \times 10^{-3}$ ; the lattice’s advantage is large  $N$  at small  $k$ . Where accuracy demands exceed the achievable rank- $k$  error, GHK’s arbitrary- $\Sigma$  generality is the right tool.

**Substitution: a  $2 \times 2$  factorial.** The design varies factor structure and noise family one axis at a time, holding the idiosyncratic scale common — coupling either axis with alternative-specific scales would confound the comparison (an alternative-specific-scale Gumbel race is not mixed logit, and heteroskedasticity alone already breaks IIA). A separate heteroskedasticity axis is left to future replication.

Ground truth is misspecified for every candidate:  $t(5)$  factors and skew-normal idiosyncratic noise standardized to mean zero and unit variance, homoskedastic. (The race is computed min-wins, so positive skew in race coordinates is *left*-skew in utility coordinates.) The four candidates hold the idiosyncratic scale fixed at one and vary  $\{\text{Gumbel, Gaussian}\} \text{ base} \times \{V = 0, V = V^*\}$ : independent Luce (= plain multinomial logit exactly), independent probit, factor mixed logit (common-scale), and factor probit. All are calibrated to the same full-menu shares (residuals  $\leq 4 \times 10^{-10}$ ).

Deletion truths come from  $2 \times 10^7$  *common* utility draws with the top three alternatives retained per draw, from which every single-deletion and pair-deletion truth follows with common random numbers: 105 informative blocks stratified by deleted mass (26 / 29 / 28 blocks at  $>10\%$  / 2–10% / 0.5–2%). The score per deletion block  $B$  is the misallocated fraction of the redistributed mass,

$$\text{TV}\left(q_{\text{model}}^{(-B)}, q_{\text{truth}}^{(-B)}\right) / \sum_{i \in B} p_i,$$

averaged within strata:

|                                  | mass $> 10\%$ | 2–10%        | 0.5–2%       |
|----------------------------------|---------------|--------------|--------------|
| independent Luce (= plain logit) | 0.229         | 0.245        | 0.261        |
| independent probit               | 0.210         | 0.232        | 0.230        |
| factor mixed logit               | 0.114         | 0.146        | 0.199        |
| factor probit                    | <b>0.045</b>  | <b>0.062</b> | <b>0.094</b> |

The factorial decomposition (top two strata): the factor increment is dominant in both families (+0.115/+0.099 within Gumbel, +0.165/+0.170 within Gaussian); the family increment alone is small (+0.019/ + 0.013 at  $V = 0$ ) but grows with factors present (+0.069/ + 0.084), a negative interaction (−0.050/ − 0.071): factor structure and the Gaussian base are complements in this design. Caveats: a single truth design and seed, and a skew-normal truth that is Gaussian-adjacent. The figure shows every block individually.

## 4 Related work

GHK and simulated likelihood are standard [2, 3, 4]; analytic-approximation lines include MACML [7]; modern Gaussian-probability methods include minimax tilting [9]. Factor dimension reduction for probit is classical: Butler–Moffitt one-factor quadrature [5], Elrod–Keane factor-analytic probit [6], and the probit probability approximations surveyed in [8]. The lattice survival-product identity and its inverse transform are from [10]. On inversion, the social-surplus identity  $p = \nabla G$  and convex-conjugate formulation of the inverse are established discrete-choice

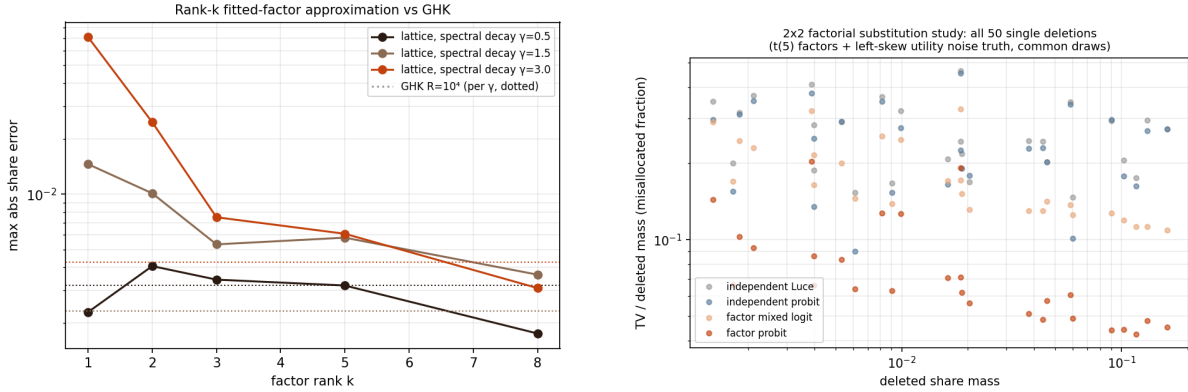


Figure 3: Left: rank- $k$  contrast-space approximation error under spectral decay  $\gamma$  (dotted: GHK at  $R = 10^4$ , per  $\gamma$ ; not wall-time matched). Right: the  $2 \times 2$  factorial substitution study — every informative single-deletion block shown individually.

results — Chiong, Galichon and Shum [15] formulate the inverse through the convex conjugate, and Norets and Takahashi [16] establish surjectivity — and factor-structured MNP for large choice sets has been pursued via Bayesian estimation by Loaiza-Maya and Nibbling [17]. The novelty claimed here is accordingly *not* well-posedness of inversion, nor dimensional reduction, both old: it is the scalable deterministic realization of the forward and inverse maps — all-alternatives assembly in  $O(QNL)$ , the  $O(QNL)$  Jacobian–vector product, and shared-field deletion counterfactuals in  $O(QN^2L)$ . Share inversion and its acceleration are central to demand estimation [11, 12]; automatic differentiation of logit-based BLP is recent [13]; a 2026 preprint approximates correlated-choice probabilities with equivariant neural networks [14], evidence of current demand for this computation. We are unaware of an automatically differentiable, large-choice-set factor-probit demand implementation built on deterministic share inversion; that precise object is what Section 5 proposes.

## 5 Discussion

Estimation is the natural next step: with the reduced coordinates of Proposition 3, gradients of an outer likelihood or GMM objective pass through the share-inversion fixed point by implicit differentiation, a candidate route to differentiable probit demand estimation at large  $N$ . Estimating  $(V, D)$  rather than supplying them raises the usual identification cautions: utility scale normalization, factor rotation, choice-invariant common loading components, and the fact that only difference covariances are identified. Beyond the Gaussian family, the same machinery computes races with arbitrary alternative-specific idiosyncratic densities; a *common-scale* Gumbel base with zero loadings reproduces the Luce model exactly (alternative-specific scales break Luce, as they should), yielding correlated-softmax generalizations. Finally, the inversion inherits the original transform’s interpolation structure (per-node location-to-probability curves are cross-correlations, evaluable at all offsets simultaneously), which we have not yet exploited.

## References

- [1] L. L. Thurstone. A law of comparative judgment. *Psychological Review* 34 (1927) 273–286.
- [2] J. Geweke, M. Keane, D. Runkle. Alternative computational approaches to inference in the multinomial probit model. *Review of Economics and Statistics* 76 (1994) 609–632.
- [3] V. Hajivassiliou, D. McFadden. The method of simulated scores for the estimation of LDV models. *Econometrica* 66 (1998) 863–896.
- [4] K. Train. *Discrete Choice Methods with Simulation*, 2nd ed. Cambridge, 2009.
- [5] J. S. Butler, R. Moffitt. A computationally efficient quadrature procedure for the one-factor multinomial probit model. *Econometrica* 50 (1982) 761–764.
- [6] T. Elrod, M. P. Keane. A factor-analytic probit model for representing the market structure in panel data. *Journal of Marketing Research* 32 (1995) 1–16.
- [7] C. Bhat. The maximum approximate composite marginal likelihood (MACML) estimation of multinomial probit-based unordered response choice models. *Transportation Research B* 45 (2011) 923–939.
- [8] R. Connors, S. Hess, A. Daly. Analytic approximations for computing probit choice probabilities. *Transportmetrica A* 10 (2014) 119–139.
- [9] Z. Botev. The normal law under linear restrictions: simulation and estimation via minimax tilting. *JRSS-B* 79 (2017) 125–148.
- [10] P. Cotton. Inferring relative ability from winning probability in multientrant contests. *SIAM J. Financial Mathematics* 12 (2021) 295–317.
- [11] S. Berry, J. Levinsohn, A. Pakes. Automobile prices in market equilibrium. *Econometrica* 63 (1995) 841–890.
- [12] C. Conlon, J. Gortmaker. Best practices for differentiated products demand estimation with pyblp. *RAND Journal of Economics* 51 (2020) 1108–1161.
- [13] A. Chia. Automatically differentiable random coefficient logistic demand estimation. arXiv:2106.04636 (2021).
- [14] E. Huch, M. Keane. Amortized inference for correlated discrete choice models via equivariant neural networks. arXiv:2603.24705 (2026).
- [15] K. Chiong, A. Galichon, M. Shum. Duality in dynamic discrete-choice models. *Quantitative Economics* 7 (2016) 83–115.
- [16] A. Norets, S. Takahashi. On the surjectivity of the mapping between utilities and choice probabilities. *Quantitative Economics* 4 (2013) 149–155.
- [17] R. Loaiza-Maya, D. Nibbering. Scalable Bayesian estimation in the multinomial probit model. *Journal of Business & Economic Statistics* 40 (2022) 1678–1690.
- [18] The `allocation` package: Thurstone-tilt portfolio construction. <https://github.com/microprediction/allocation>.